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MICROELECTRONIC ASSEMBLIES HAVING A HYBRID BONDED INTERPOSER FOR DIE-TO-DIE FAN-OUT SCALING

Abstract

Microelectronic assemblies, and related devices and methods, are disclosed herein. In some embodiments, a microelectronic assembly may include an interposer having a first surface and an opposing second surface; a first die coupled to the first surface of the interposer by a first hybrid bonding region having a first pitch; a second die coupled to the second surface of the interposer by a second hybrid bonding region having a second pitch different from the first pitch.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of (and claims the benefit and priority under 35 U.S.C. 120 of) U.S. patent application Ser. No. 17/347,394, filed Jun. 14, 2021, entitled “MICROELECTRONIC ASSEMBLIES HAVING A HYBRID BONDED INTERPOSER FOR DIE-TO-DIE FAN-OUT SCALING,” the disclosure of which is considered part of, and is incorporated by reference in, the disclosure of this application.

BACKGROUND

[0002] Integrated circuit (IC) dies are coupled to each other via 3-dimensional (3D) stacking for improved interconnect pitch and reduced overall surface area of an IC package. IC dies for 3D stacking are designed to have a same pitch to ensure compatibility.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Embodiments described herein illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like references indicate similar features. The following figures are illustrative, and other processing techniques or stages can be used in accordance with the subject matter described herein. Furthermore, some conventional details have been omitted so as not to obscure from the inventive concepts described herein.

[0004] FIG. 1A is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

[0005] FIG. 1B is a side, cross-sectional exploded view of a portion of the microelectronic assembly of FIG. 1A, in accordance with various embodiments.

[0006] FIGS. 2A-2C are side, cross-sectional views of example microelectronic assemblies, in accordance with various embodiments.

[0007] FIG. 3 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments.

[0008] FIGS. 4A and 4B are side, cross-sectional views of example microelectronic assemblies, in accordance with various embodiments.

[0009] FIGS. 5A-5F are side, cross-sectional views of various stages in an example process for manufacturing a microelectronic assembly, in accordance with various embodiments.

[0010] FIGS. 6A-6E are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 1A, in accordance with various embodiments.

[0011] FIGS. 7A-7E are side, cross-sectional views of various stages in an example process for manufacturing a microelectronic assembly, in accordance with various embodiments.

[0012] FIG. 8 is a top view of a wafer and dies that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

[0013] FIG. 9 is a cross-sectional side view of an IC device that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

[0014] FIG. 10 is a cross-sectional side view of an IC device assembly that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

[0015] FIG. 11 is a block diagram of an example electrical device that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

DETAILED DESCRIPTION

[0016] Microelectronic assemblies, and related devices and methods, are disclosed herein. For example, in some embodiments, a microelectronic assembly may include a die-level interposer having a first surface and an opposing second surface; a first die coupled to the first surface of the die-level interposer by a first hybrid bonding region having a first pitch; and a second die coupled to the second surface of the die-level interposer by a second hybrid bonding region having a second pitch different from the first pitch.

[0017] Communicating large numbers of signals between two or more dies in a multi-die IC package is challenging due to the increasingly small size of such dies and increased use of stacking dies. This becomes more difficult for stacking of dies having different pitches. Conventional approaches include designing each multi-die IC package to have a same pitch. Other conventional approaches include designing each multi-die IC package to account for any differences in pitch. Although all combinations of stacked dies may be pre-designed, the large number of possible combinations is time and cost prohibitive and the ability to incorporate third-party dies is limited. Various ones of the microelectronic assemblies disclosed herein may more efficiently use design resources relative to conventional approaches by reducing the need to pre-design each multi-die IC package by providing a die-level interposer that enables fan-out scaling for hybrid bonded stacked dies with different pitches. The microelectronic assemblies disclosed herein may be particularly advantageous for small and low-profile applications in computers, tablets, industrial robots, and consumer electronics (e.g., wearable devices).

[0018] In the following detailed description, reference is made to the accompanying drawings that form a part hereof wherein like numerals designate like parts throughout, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense.

[0019] Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the claimed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed, and/or described operations may be omitted in additional embodiments.

[0020] For the purposes of the present disclosure, the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The drawings are not necessarily to scale. Although many of the drawings illustrate rectilinear structures with flat walls and right-angle corners, this is simply for ease of illustration, and actual devices made using these techniques will exhibit rounded corners, surface roughness, and other features.

[0021] The description uses the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. Furthermore, the terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. When used to describe a range of dimensions, the phrase “between X and Y” represents a range that includes X and Y. The terms “top,” “bottom,” etc. may be used herein to explain various features of the drawings, but these terms are simply for ease of discussion, and do not imply a desired or required orientation. As used herein, the term “thickness” refers to a

dimension of a certain element or layer as measured along the z-axis, the term “width” refers to a dimension of a certain element or layer as measured along the y-axis, while the term “length” refers to a dimension of a certain element or layer as measured along the x-axis. Although certain elements may be referred to in the singular herein, such elements may include multiple sub-elements. For example, “a dielectric material” may include one or more dielectric materials. Throughout the specification, and in the claims, the term “connected” means a direct connection, such as electrical, mechanical, or magnetic connection between the things that are connected, without any intermediary devices. The term “coupled” means a direct or indirect connection, such as a direct electrical, mechanical, or magnetic connection between the things that are connected or an indirect connection, through one or more passive or active intermediary devices. The meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.” Unless otherwise specified the use of the ordinal adjectives “first,” “second,” and “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to, and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner. For ease of discussion, the drawings of FIGS. 1A and 1B may be referred to herein as “FIG. 1,” and the drawings of FIGS. 2A-2C may be referred to herein as “FIG. 2,” etc.

[0022] FIG. 1 is a side, cross-sectional view of a microelectronic assembly **100**, in accordance with various embodiments. The microelectronic assembly **100** may include a plurality of dies having different pitches coupled via hybrid bonding to a die-level interposer **104** for die-to-die fan-out scaling. In particular, the microelectronic assembly **100** may include a first die **114-1** with a first pitch coupled to a first surface **170-1** of the die-level interposer **104** by a first hybrid bonding (HB) region **130-1** and a second die **114-2** with a second pitch different from the first pitch coupled to an opposing second surface **170-2** of the die-level interposer **104** by a second HB region **130-2**. The microelectronic assembly **100** may further include a third die **114-3** with a third pitch different from the first and second pitches coupled to the second surface **170-2** of the die-level interposer **104** by a third HB region **130-3**. The bottom surface of the first die **114-1** may further be coupled to a substrate **150** by a fourth HB region **130-4** (e.g., the die **114-1** may include TSVs coupled to the bottom surface of the die for forming connections between the substrate **150** and the die level interposer **104**). In some embodiments, the bottom surface of the first die **114-1** may be coupled to the substrate **150** by non-hybrid bonded interconnects (not shown), such as solder (e.g., as described below with reference to solder **120**), or metal-to-metal contacts, among others.

[0023] The die-level interposer **104** may include any suitable number of layers and any suitable arrangement of layers that form a pitch translation structure, including a standoff layer **101** (e.g., a thick substrate layer) and, in some embodiments, a routing layer **148** (e.g., a thin metallization layer). For example, as shown in FIG. 1A, the die-level interposer **104** may include a routing layer **148** at the first surface **170-1** and a standoff layer **101** on the routing layer **148** (e.g., at the second surface **170-2**). The routing layer **148** may be formed to provide a dense fan-out structure for translating a smaller pitch (e.g., a pitch of the first die) to a larger pitch (e.g., a pitch of the second and third dies **114-2**, **114-3**) within a lesser thickness. The standoff layer **101** may be formed to provide a less dense fan-out structure for translating a smaller pitch (e.g., a pitch of the first die) to a larger pitch (e.g., a pitch of the second and third dies **114-2**, **114-3**) over a greater thickness. In some embodiments, the standoff layer **101** may be formed to provide distance or spacing (e.g., z-height) between the first die **114-1** and the second **114-2** and/or the third die **114-3** (e.g., to create distance from a radio frequency source).

[0024] The routing layer **148** may include an insulating material **105** (e.g., one or more dielectric materials formed in multiple layers, as known in the art) and one or more conductive pathways **111** through the insulating material **105** (e.g., including conductive lines and/or conductive vias, as shown). The insulating material **105** may include an inorganic dielectric material, such as silicon and nitrogen (e.g., in the form of silicon nitride); silicon and oxygen (e.g., in the form of silicon

oxide); silicon and carbon (e.g., in the form of silicon carbide); silicon, carbon, and oxygen (e.g., in the form of silicon oxycarbide); silicon, carbon, and nitrogen (e.g., in the form of silicon carbonitride); carbon and oxygen (e.g., in the form of a carbon-doped oxide); silicon, oxygen, and nitrogen (e.g., in the form of silicon oxynitride); or silicon, oxygen, carbon, and hydrogen (e.g., in the form of tetraethyl orthosilicate (TEOS)); and combinations thereof. In some embodiments, the insulating material **105** may include fused silica, glass, glass-reinforced epoxy matrix materials, alkali-free borosilicate, or a low-k or ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, organic polymeric dielectrics, photo-imageable dielectrics, and/or benzocyclobutene-based polymers). In some embodiments, the insulating material **105** may include an organic dielectric, such as polyimide or polybenzoxazole, or may include an organic polymer matrix (e.g., epoxide) with a filler material (which may be inorganic, such as silicon nitride, silicon oxide, or aluminum oxide). In some embodiments, the insulating material **105** may include an organic material coated by an inorganic material (e.g., an organic material coated by silicon nitride). In some embodiments, the insulating material **105** may include a ceramic. The conductive pathways **111** may be formed of any suitable conductive material, such as copper, silver, nickel, gold, aluminum, other metals or alloys, or combinations of materials, for example. In some embodiments, the routing layer **148** may be fabricated on the standoff layer **101** using back-end-of-line (BEOL) processes, as known in the art, prior to coupling by HB region **130-1** to the first die **114-1**, as described below with reference to FIG. **6**. In some embodiments, the routing layer **148** has a thickness (e.g., z-height) between 0.05 microns and 5 microns. In some embodiments, the routing layer **148** has a thickness between 5 microns and 100 microns (e.g., between 5 microns and 50 microns, between 25 microns and 75 microns, and between 50 microns and 100 microns).

[0025] The standoff layer **101** may include an insulating material **107** (e.g., one or more dielectric materials formed in multiple layers, as known in the art) and one or more through-substrate vias (TSVs) **113** through the insulating material **107**, as shown. The insulating material of the standoff layer **101** may include silicon and oxygen (e.g., in the form of silicon oxide), silicon and carbon (e.g., in the form of silicon carbide), oxynitride, polyimide materials, fused silica, glass, glass-reinforced epoxy matrix materials, or a low-k or ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, organic polymeric dielectrics, photo-imageable dielectrics, and/or benzocyclobutene-based polymers). In some embodiments, the insulating material of a standoff layer may include a semiconductor material, such as silicon, and one or more additional materials. In some embodiments, the insulating material **107** may include an organic dielectric, such as polyimide or polybenzoxazole, or may include an organic polymer matrix (e.g., epoxide) with a filler material (which may be inorganic, such as silicon nitride, silicon oxide, or aluminum oxide). In some embodiments, the insulating material **105** may include an organic material coated by an inorganic material. The TSVs **113** may be formed of any suitable conductive material, such as copper, silver, nickel, gold, aluminum, other metals or alloys, or combinations of materials, for example. In some embodiments, the standoff layer **101** may be wafer (e.g., wafer **1500** of FIG. **8**) including TSVs **113**. In some embodiments, the standoff layer **101** may be wafer (e.g., wafer **1500** of FIG. **8**) including TSVs and a top dielectric and metallization layer for fabricating a routing layer **148**, as described below with reference to FIG. **6**. In some embodiments, the standoff layer **101** has a thickness (e.g., z-height) between 5 microns and 400 microns. In some embodiments, the TSVs **113** have an aspect ratio between 40:1 and 20:1. The TSVs **113** may have any suitable shapes, sizes, and arrangements. In some particular embodiments, TSVs **113** may have footprints shaped as convex polygons (e.g., squares, rectangles, octagons, cross shapes, etc.) or circular (e.g., ovals, circles, etc.).

[0026] The microelectronic assembly **100** may further include a mold material **126**, a support component **182**, an underfill material **138**, a heat transfer structure **156**, and a thermal interface material (TIM) **154**. A number of elements are illustrated in FIG. **1** as included in the

microelectronic assembly **100**, but a number of these elements may not be present in a microelectronic assembly **100**. For example, in various embodiments, the substrate **150**, the mold material **126**, the underfill material **138**, the support component **182**, the underfill material **138**, the heat transfer structure **156**, and/or the thermal interface material (TIM) **154** may not be included. Further, FIG. **1** illustrates a number of elements that are omitted from subsequent drawings for ease of illustration, but may be included in any of the microelectronic assemblies **100** disclosed herein. Examples of such elements include the substrate **150**, the mold material **126**, the underfill material **138**, the support component **182**, the underfill material **138**, the heat transfer structure **156**, and/or the thermal interface material (TIM) **154**. Many of the elements of the microelectronic assembly **100** of FIG. **1** are included in other ones of the accompanying drawings; the discussion of these elements is not repeated when discussing these drawings, and any of these elements may take any of the forms disclosed herein. In some embodiments, individual ones of the microelectronic assemblies **100** disclosed herein may serve as a system-in-package (SiP) in which multiple die **114** having different functionality are included. In such embodiments, the microelectronic assembly **100** may be referred to as an SiP.

[0027] The microelectronic assembly **100** may include a second die **114-2** coupled to a die-level interposer **104** by a HB (HB) region **130-2**. In particular, as illustrated in FIG. **1B**, the HB region **130-2** may include a HB interface **180-1A** at the top surface of the die-level interposer **104**, with the HB interface **180-1A** including a set of conductive HB contacts **110** and a HB dielectric **108** around the HB contacts **110** of the HB interface **180-1A**. The HB region **130-2** may also include a HB interface **180-1B** at the bottom surface of the die **114-2**, with the HB interface **180-1B** including a set of HB contacts **110** and a HB dielectric **108** around the HB contacts **110** of the HB interface **180-1B**. The HB contacts **110** of the HB interface **180-1A** of die-level interposer **104** may align with the HB contacts **110** of the HB interface **180-1B** of the die **114-2** so that, in the microelectronic assembly **100**, the HB contacts **110** of the die **114-2** are in contact with the HB contacts **110** of the die-level interposer **104**. In the microelectronic assembly **100** of FIG. **1**, the HB interface **180-1A** of the die-level interposer **104** may be bonded (e.g., electrically and mechanically) with the HB interface **180-1B** of the die **114-2** to form the HB region **130-2** coupling the die-level interposer **104** and the die **114-2**.

[0028] The microelectronic assembly **100** may further include a third die **114-3** coupled to a die-level interposer **104** by a HB region **130-3**. In particular, as illustrated in FIG. **1B**, the HB region **130-3** may include a HB interface **180-2A** at the top surface of the die-level interposer **104**, with the HB interface **180-2A** including a set of conductive HB contacts **110** and a HB dielectric **108** around the HB contacts **110** of the HB interface **180-2A**. The HB region **130-3** may also include a HB interface **180-2B** at the bottom surface of the die **114-3**, with the HB interface **180-2B** including a set of HB contacts **110** and a HB dielectric **108** around the HB contacts **110** of the HB interface **180-2B**. The HB contacts **110** of the HB interface **180-2A** of the die-level interposer **104** may align with the HB contacts **110** of the HB interface **180-2B** of the die **114-3** so that, in the microelectronic assembly **100**, the HB contacts **110** of the die **114-3** are in contact with the HB contacts **110** of the die-level interposer **104**. In the microelectronic assembly **100** of FIG. **1**, the HB interface **180-2A** of the die-level interposer **104** may be bonded (e.g., electrically and mechanically) with the HB interface **180-2B** of the die **114-3** to form the HB region **130-3** coupling the die-level interposer **104** and the die **114-3**. The microelectronic assembly **100** may further include the first die **114-1** coupled on a top surface to a die-level interposer **104** by a HB region **130-1** and on a bottom surface to a substrate **150** by a HB region **130-4**, where the HB regions **130-1**, **130-4** include HB interfaces **180** (not labeled). More generally, the HB regions **130** disclosed herein may include two complementary HB interfaces **180** bonded together; for ease of illustration, many of the subsequent drawings may omit the identification of the HB interfaces **180** to improve the clarity of the drawings.

[0029] As used herein, the term “hybrid bonding” is used to include techniques in which the HB

dielectric **108** of opposing HB interfaces **180** are brought into contact first, then subject to heat and sometimes compression, or techniques in which the HB contacts **110** and the HB dielectric **108** of opposing HB interfaces **180** are brought into contact substantially simultaneously, then subject to heat and compression. In such techniques, the HB contacts **110** and the HB dielectric **108** at one HB interface **180** are brought into contact with the HB contacts **110** and the HB dielectric **108** at another HB interface **180**, respectively, and elevated pressures and/or temperatures may be applied to cause the contacting HB contacts **110** and/or the contacting HB dielectrics **108** to bond. In some embodiments, this bond may be achieved without the use of intervening solder or an anisotropic conductive material, while in some other embodiments, a thin cap of conductive material (such as Au, Ti, Manganin alloy, etc.) may be used in a HB interconnect to prevent oxidation of the pad surface or to improve the metal to metal bond, and this conductive material may become an intermetallic compound (IMC) in the HB region **130** during processing. HB interconnects may be capable of reliably conducting a higher current than other types of interconnects; for example, some conventional solder interconnects may form large volumes of brittle IMCs when current flows, and the maximum current provided through such interconnects may be constrained to mitigate mechanical failure. Although FIGS. **1A** and **1B** show the HB dielectric **108** as extending fully along the entire top surface of the substrate **150**, in some embodiments, the HB dielectric **108** may extend only along a portion of the top surface of the substrate **150**, for example, when the substrate **150** has a larger surface area as compared to the die **114-1**, such that the HB dielectric **108** is only within the HB region **130**.

[0030] A HB dielectric **108** may include one or more dielectric materials, such as one or more inorganic dielectric materials. For example, a HB dielectric **108** may include silicon and nitrogen (e.g., in the form of silicon nitride); silicon and oxygen (e.g., in the form of silicon oxide); silicon, carbon, and nitrogen (e.g., in the form of silicon carbonitride); carbon and oxygen (e.g., in the form of a carbon-doped oxide); silicon, oxygen, and nitrogen (e.g., in the form of silicon oxynitride); aluminum and oxygen (e.g., in the form of aluminum oxide); titanium and oxygen (e.g., in the form of titanium oxide); hafnium and oxygen (e.g., in the form of hafnium oxide); silicon, oxygen, carbon, and hydrogen (e.g., in the form of tetraethyl orthosilicate (TEOS)); zirconium and oxygen (e.g., in the form of zirconium oxide); niobium and oxygen (e.g., in the form of niobium oxide); tantalum and oxygen (e.g., in the form of tantalum oxide); and combinations thereof.

[0031] A HB contact **110** may include a pillar, a pad, or other structure. The HB contacts **110**, although depicted in the accompanying drawings in the same manner at both HB interfaces **180** of a HB region **130**, may have a same structure at both HB interfaces **180**, or the HB contacts **110** at different HB interfaces **180** may have different structures. For example, in some embodiments, a HB contact **110** in one HB interface **180** may include a metal pillar (e.g., a copper pillar), and a complementary HB contact **110** in a complementary HB interface **180** may include a metal pad (e.g., a copper pad) recessed in a dielectric. A HB contact **110** may include any one or more conductive materials, such as copper, manganese, titanium, gold, silver, palladium, nickel, copper and aluminum (e.g., in the form of a copper aluminum alloy), tantalum (e.g., tantalum metal, or tantalum and nitrogen in the form of tantalum nitride), cobalt, cobalt and iron (e.g., in the form of a cobalt iron alloy), or any alloys of any of the foregoing (e.g., copper, manganese, and nickel in the form of manganin). In some embodiments, the HB dielectric **108** and the HB contacts **110** of a HB interface **180** may be manufactured using low-temperature deposition techniques (e.g., techniques in which deposition occurs at temperatures below 250 degrees Celsius, or below 200 degrees Celsius), such as low-temperature PECVD.

[0032] FIGS. **1A** and **1B** also illustrate the die **114-1** coupled to the substrate **150** by a HB region **130-4**. Although FIG. **1** depicts a single die **114** coupled to the substrate **150** by HB region **130**, this number and arrangement are simply illustrative, and a microelectronic assembly **100** may include any desired number and arrangement of die **114** coupled to a substrate **150** and to other die **114** by HB regions **130**. Although a single reference numeral “108” is used to refer to the HB dielectrics of

multiple different HB interfaces **180** (and different HB regions **130**), this is simply for ease of illustration, and the HB dielectric **108** of different HB interfaces **180** (even within a single HB region **130**) may have different materials and/or structures. Similarly, although a single reference numeral “110” is used to refer to the HB contacts of multiple different HB interfaces **180** (and different HB regions **130**), this is simply for ease of illustration, and the HB contacts **110** of different HB interfaces **180** (even within a single HB region **130**) may have different materials and/or structures.

[0033] The substrate **150** may include an insulating material **106** (e.g., one or more dielectric materials formed in multiple layers, as known in the art) and one or more conductive pathways **112** through the insulating material **106** (e.g., including conductive lines **109** and/or conductive vias **116**, as shown). In some embodiments, the insulating material **106** of the substrate **150** includes an inorganic dielectric material, such as silicon and nitrogen (e.g., in the form of silicon nitride); silicon and oxygen (e.g., in the form of silicon oxide); silicon and carbon (e.g., in the form of silicon carbide); silicon, carbon, and oxygen (e.g., in the form of silicon oxycarbide); silicon, carbon, and nitrogen (e.g., in the form of silicon carbonitride); carbon and oxygen (e.g., in the form of a carbon-doped oxide); silicon, oxygen, and nitrogen (e.g., in the form of silicon oxynitride); or silicon, oxygen, carbon, and hydrogen (e.g., in the form of tetraethyl orthosilicate (TEOS)); and combinations thereof. In some embodiments, the insulating material **106** of the substrate **150** includes an insulating metal oxide, such as aluminum and oxygen (e.g., in the form of aluminum oxide); titanium and oxygen (e.g., in the form of titanium oxide); hafnium and oxygen (e.g., in the form of hafnium oxide); zirconium and oxygen (e.g., in the form of zirconium oxide); niobium and oxygen (e.g., in the form of niobium oxide); or tantalum and oxygen (e.g., in the form of tantalum oxide); and combinations thereof. In some embodiments, the substrate **150** may be semiconductor-based (e.g., silicon-based) or glass-based. In some embodiments, the substrate **150** is a silicon wafer or die. In some embodiments, the substrate **150** may be a silicon-on-insulator (SOI) and may further include layers of silicon and germanium (e.g., in the form of silicon germanium), gallium and nitrogen (e.g., in the form of gallium nitride), indium and phosphorous (e.g., in the form of indium phosphide), among others. In some embodiments, the insulating material **106** of the substrate **150** may be an organic material, such as polyimide or polybenzoxazole, or may include an organic polymer matrix (e.g., epoxide) with a filler material (which may be inorganic, such as silicon nitride, silicon oxide, or aluminum oxide). In some such embodiments, the substrate **150** may be referred to as an “organic interposer.” In some embodiments, the insulating material **106** of a substrate **150** may be provided in multiple layers of organic buildup film. Organic interposers **150** may be less expensive to manufacture than semiconductor- or glass-based interposers, and may have electrical performance advantages due to the low dielectric constants of organic insulating materials **106** and the thicker lines that may be used (allowing for improved power delivery, signaling, and potential thermal benefits). Organic interposers **150** may also have larger footprints than can be achieved for semiconductor-based interposers, which are limited by the size of the reticle used for patterning. Further, organic interposers **150** may be subject to less restrictive design rules than those that constrain semiconductor- or glass-based interposers, allowing for the use of design features such as non-Manhattan routing (e.g., not being restricted to using one layer for horizontal interconnects and another layer for vertical interconnects) and the avoidance of through-substrate vias (TSVs) such as through-silicon vias or through-glass vias (which may be limited in the achievable pitch, and may result in less desirable power delivery and signaling performance). Conventional integrated circuit packages including an organic interposer have been limited to solder-based attach technologies, which may have a lower limit on the achievable pitch that precludes the use of conventional solder-based interconnects to achieve the fine pitches desired for next generation devices. Utilizing an organic substrate **150** in a microelectronic assembly **100** with hybrid bonding, as disclosed herein, may leverage these advantages of organic interposers in combination with the ultra-fine pitch (e.g., the pitch **128** discussed below) achievable by hybrid

bonding (and previously only achievable when using semiconductor-based interposers), and thus may support the design and fabrication of large and sophisticated die complexes that can achieve packaged system competition performance and capabilities not enabled by conventional approaches.

[0034] In other embodiments, the insulating material **106** of the substrate **150** may include a fire retardant grade **4** material (FR-4), bismaleimide triazine (BT) resin, or low-k or ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, and porous dielectrics). When the substrate **150** is formed using standard printed circuit board (PCB) processes, the insulating material **106** may include FR-4, and the conductive pathways **112** in the substrate **150** may be formed by patterned sheets of copper separated by buildup layers of the FR-4. In some such embodiments, the substrate **150** may be referred to as a “package substrate” or a “circuit board.”

[0035] In some embodiments, one or more of the conductive pathways **112** in the substrate **150** may extend between a conductive contact at the top surface of the substrate **150** (e.g., one of the HB contacts **110**) and a conductive contact **118** at the bottom surface of the substrate **150**. In some embodiments, one or more of the conductive pathways **112** in the substrate **150** may extend between different conductive contacts at the top surface of the substrate **150** (e.g., between different HB contacts **110** potentially in different HB regions **130**). In some embodiments, one or more of the conductive pathways **112** in the substrate **150** may extend between different conductive contacts **118** at the bottom surface of the substrate **150**. In some embodiments, a substrate **150** may only include conductive pathways **112**, and may not contain active or passive circuitry. In other embodiments, a substrate **150** may include active or passive circuitry (e.g., transistors, diodes, resistors, inductors, and capacitors, among others). In some embodiments, a substrate **150** may include one or more device layers including transistors. Although FIGS. **1A** and **1B** (and others of the accompanying drawings) illustrate a specific number and arrangement of conductive pathways **112** in the substrate **150**, these are simply illustrative, and any suitable number and arrangement may be used. The conductive pathways **112** disclosed herein (e.g., including lines **109** and/or vias **116**) may be formed of any appropriate conductive material, such as copper, silver, nickel, gold, aluminum, other metals or alloys, or combinations of materials, for example.

[0036] In some embodiments, a die **114** may include an IC die (packaged or unpackaged) or a stack of an IC dies (e.g., a high-bandwidth memory dies stack). In some such embodiments, the insulating material of a die **114** may include silicon dioxide, silicon nitride, oxynitride, polyimide materials, glass-reinforced epoxy matrix materials, or a low-k or ultra low-k dielectric (e.g., carbon-doped dielectrics, fluorine-doped dielectrics, porous dielectrics, organic polymeric dielectrics, photo-imageable dielectrics, and/or benzocyclobutene-based polymers). In some further embodiments, the insulating material of a die **114** may include a semiconductor material, such as silicon, germanium, or a III-V material (e.g., gallium nitride), and one or more additional materials. For example, an insulating material of a die **114** may include silicon oxide or silicon nitride. The conductive pathways in a die **114** may include conductive lines and/or conductive vias, and may connect any of the conductive contacts in the die **114** in any suitable manner (e.g., connecting multiple conductive contacts on a same surface or on different surfaces of the die **114**). The die **114** may include TSVs (not shown). Example structures that may be included in the die **114** disclosed herein are discussed below with reference to FIG. **9**. In particular, a die **114** may include active and/or passive circuitry (e.g., transistors, diodes, resistors, inductors, and capacitors, among others). In some embodiments, a die **114** may include one or more device layers including transistors. When a die **114** includes active circuitry, power and/or ground signals may be routed through the substrate **150** and to/from a die **114** through a HB region **130** (and further through intervening die **114**). The die **114** of the microelectronic assembly **100** of FIG. **1** may be single-sided components, in the sense that an individual die **114** only have conductive contacts (e.g., HB contacts **110**) on a single surface of the individual die **114** (e.g., die **114-2**, **114-3**), or may be double-sided (or “multi-level,” or “omni-directional”) components with conductive contacts on

multiple surfaces of the component (e.g., die **114-1**).

[0037] Additional components (not shown), such as surface-mount resistors, capacitors, and/or inductors, may be disposed on the top surface or the bottom surface of the substrate **150**, or embedded in the substrate **150**. The microelectronic assembly **100** of FIG. **1** also includes a support component **182** coupled to the substrate **150**. In the particular embodiment of FIG. **1**, the support component **182** includes conductive contacts **118** that are electrically coupled to complementary conductive contacts **118** of the substrate **150** by intervening solder **120** (e.g., solder balls in a ball grid array (BGA) arrangement), but any suitable interconnect structures may be used (e.g., pins in a pin grid array arrangement, lands in a land grid array arrangement, pillars, pads and pillars, etc.). The solder **120** utilized in the microelectronic assemblies **100** disclosed herein may include any suitable materials, such as lead/tin, tin/bismuth, eutectic tin/silver, ternary tin/silver/copper, eutectic tin/copper, tin/nickel/copper, tin/bismuth/copper, tin/indium/copper, tin/zinc/indium/bismuth, or other alloys. In some embodiments, the couplings between the substrate **150** and the support component **182** may be referred to as second-level interconnects (SLI) or multi-level interconnects (MLI).

[0038] In some embodiments, the support component **182** may be a package substrate (e.g., may be manufactured using PCB processes, as discussed above). In some embodiments, the support component **182** may be a circuit board (e.g., a motherboard), and may have other components attached to it (not shown). The support component **182** may include conductive pathways and other conductive contacts (not shown) for routing power, ground, and signals through the support component **182**, as known in the art. In some embodiments, the support component **182** may include another IC package, an interposer, or any other suitable component. An underfill material **138** may be disposed around the solder **120** coupling the substrate **150** to the support component **182**. In some embodiments, the underfill material **138** may include an epoxy material.

[0039] In some embodiments, the support component **182** may be a lower density component, while the substrate **150** and/or the die **114** may be higher density components. As used herein, the term “lower density” and “higher density” are relative terms indicating that the conductive pathways (e.g., including conductive lines and conductive vias) in a lower density component are larger and/or have a greater pitch than the conductive pathways in a higher density component. In some embodiments, a die **114** may be a higher density component, and a substrate **150** may be a lower density component. In some embodiments, a higher density component may be manufactured using a dual damascene or single damascene process (e.g., when the higher density component is a die), while a lower density component may be manufactured using a semi-additive or modified semi-additive process (with small vertical interconnect features formed by advanced laser or lithography processes) (e.g., when the lower density component is a package substrate or an interposer). In some other embodiments, a higher density component may be manufactured using a semi-additive or modified semi-additive process (e.g., when the higher density component is a package substrate or an interposer), while a lower density component may be manufactured using a semi-additive or a subtractive process (using etch chemistry to remove areas of unwanted metal, and with coarse vertical interconnect features formed by a standard laser process) (e.g., when the lower density component is a PCB).

[0040] The microelectronic assembly **100** of FIG. **1** may also include a mold material **126**. The mold material **126** may extend around one or more of the die **114** on the die-level interposer **104** as well as around one or more of the die **114** on the substrate **150** (not shown). In some embodiments, the mold material **126** may extend between multiple die **114** on the die-level interposer **104** and around the HB regions **130**. In some embodiments, the mold material **126** may extend around one or more of the die **114** on the substrate **150** (not shown). The mold material **126** may be an insulating material, such as an appropriate epoxy material. The mold material **126** may be selected to have a coefficient of thermal expansion (CTE) that may mitigate or minimize the stress between the die **114** and the die-level interposer **104** and/or the substrate **150** arising from uneven thermal

expansion in the microelectronic assembly **100**. In some embodiments, the CTE of the mold material **126** may have a value that is intermediate to the CTE of the die-level interposer **104** (e.g., the CTE of the insulating material of the die-level interposer **104**) and a CTE of the die **114**. In some embodiments, the mold material **126** used in a microelectronic assembly **100** may be selected at least in part for its thermal properties. For example, one or more mold materials **126** used in a microelectronic assembly **100** may have low thermal conductivity (e.g., conventional mold compounds) to retard heat transfer, or may have high thermal conductivity (e.g., mold materials including metal or ceramic particles with high thermal conductivity, such as copper, silver, diamond, silicon carbide, aluminum nitride, and boron nitride, among others) to facilitate heat transfer. Any of the mold materials **126** referred to herein may include one or more different materials with different material compositions.

[0041] The microelectronic assembly **100** of FIG. **1** may also include a TIM **154**. The TIM **154** may include a thermally conductive material (e.g., metal particles) in a polymer or other binder. The TIM **154** may be a thermal interface material paste or a thermally conductive epoxy (which may be a fluid when applied and may harden upon curing, as known in the art). The TIM **154** may provide a path for heat generated by the die **114** to readily flow to the heat transfer structure **156**, where it may be spread and/or dissipated. Some embodiments of the microelectronic assembly **100** of FIG. **1** may include a sputtered metallization (not shown) across the top surfaces of the mold material **126** and the die **114**; the TIM **154** (e.g., a solder TIM) may be disposed on this metallization.

[0042] The microelectronic assembly **100** of FIG. **1** may also include a heat transfer structure **156**. The heat transfer structure **156** may be used to move heat away from one or more of the die **114** (e.g., so that the heat may be more readily dissipated). The heat transfer structure **156** may include any suitable thermally conductive material (e.g., metal, appropriate ceramics, etc.), and may include any suitable features (e.g., a heat spreader, a heat sink including fins, a cold plate, etc.). In some embodiments, a heat transfer structure **156** may be or may include an integrated heat spreader (IHS).

[0043] The elements of a microelectronic assembly **100** may have any suitable dimensions. Only a subset of the accompanying drawings are labeled with reference numerals representing dimensions, but this is simply for clarity of illustration, and any of the microelectronic assemblies **100** disclosed herein may have components having the dimensions discussed herein. In some embodiments, the thickness **184** of the substrate **150** may be between 20 microns and 200 microns. In some embodiments, the thickness **188** of a HB region **130** may be between 50 nanometers and 5 microns. In some embodiments, a thickness **190** of a die **114** may be between 5 microns and 800 microns. In some embodiments, a thickness **191** of a die-level interposer **104** may be between 5 microns and 1000 microns (e.g., between 5 microns and 400 microns, between 400 microns and 800 microns, or between 400 microns and 1000 microns). In some embodiments, a pitch **128** of the HB contacts **110** in a HB region **130-1** may be less than 10 microns (e.g., between 0.1 microns and 10 microns, or between 1 micron and 9 microns). In some embodiments, a pitch **128** of the HB contacts **110** in a HB regions **130-2**, **130-3** may be between 10 microns and 200 microns. In some embodiments, a pitch **128** of the HB contacts **110** in a HB region **130-4** may be less than 20 microns (e.g., between 0.1 microns and 20 microns).

[0044] FIGS. **2A-2C** are side, cross-sectional views of example microelectronic assemblies, in accordance with various embodiments. FIGS. **2A-2C** are example microelectronic assemblies **100** including a die-level interposer **104** having a first standoff layer **101-1** and a second standoff layer **101-2**. As shown in FIGS. **2A-2C**, the microelectronic assembly **100** may include a die-level interposer **104** having a first standoff layer **101-1** at a first surface **170-1** and a second standoff layer **101-2** on the first standoff layer **101-1** and coupled to the first standoff layer **101-1** by a HB region **130-5**. As shown in FIGS. **2A** and **2B**, the microelectronic assembly **100** may further include a first die **114-1** with a first pitch coupled to a first surface **170-1** of the die-level interposer **104** by

a first hybrid bonding (HB) region **130-1** and a second die **114-2** with a second pitch different from the first pitch coupled to an opposing second surface **170-2** of the die-level interposer **104** by a second HB region **130-2**. The microelectronic assembly **100** may further include a third die **114-3** with a third pitch different from the first and second pitches coupled to the second surface **170-2** of the die-level interposer **104** by a third HB region **130-3**. The first and second standoff layers **101-1**, **101-2** may include TSVs **113** that provide pitch translation between the first pitch of the first die **114-1** and the second and third pitches of the second and third dies **114-2**, **114-3**, respectively. As shown in FIG. 2A, the TSVs **113** may provide vertical pitch translation between the first and second standoff layers **101-1**, **101-2**, where a same number of TSVs in the first standoff layer **101-1** are coupled to a same number of TSVs in the second standoff layer **101-2** (e.g., one TSV in the first standoff layer **101-1** to one TSV in the second standoff layer **101-2**). As further shown in FIG. 2A, the TSVs **113** may provide vertical pitch translation between the first and second standoff layers **101-1**, **101-2** at the HB region **130-5** and at the second and third HB regions **130-2**, **130-3**, where a plurality of TSVs **113** are coupled to a single HB contact **110**. In some embodiments, a greater number of TSVs **113** in the first standoff layer **101-1** are coupled to a smaller number of TSVs in the second standoff layer **101-2** via a HB contact **110** in the HB region **130-5** (e.g., two TSVs in the first standoff layer **101-1** to three TSVs in the second standoff layer **101-2**). In some embodiments, a smaller number of TSVs **113** in the first standoff layer **101-1** are coupled to a greater number of TSVs **113** in the second standoff layer **101-2** via a HB contact **110** in the HB region **130-5** (e.g., one TSV in the first standoff layer **101-1** to two TSVs in the second standoff layer **101-2**). In some embodiments, a same number of TSVs in the first standoff layer **101-1** are coupled to a same number of TSVs in the second standoff layer **101-2** (e.g., one TSV in the first standoff layer **101-1** to one TSV in the second standoff layer **101-2**). As shown in FIG. 2B, the TSVs **113** may provide vertical and lateral pitch translation between the first and second standoff layers **101-1**, **101-2** by laterally coupling **117** a plurality of TSVs at the HB region **130-5** (e.g., a TSV **113** within a footprint of the second die **114-2** is coupled to the third die **114-3** via the HB region **130-5** and the third HB region **130-3**). In FIG. 2C, the microelectronic assembly **100** may include a first die **114-1** with a first pitch coupled to a first surface **170-1** of the die-level interposer **104** by a first HB region **130-1**, a second die **114-2** with a second pitch different from the first pitch coupled to an opposing second surface **170-2** of the die-level interposer **104** by a second HB region **130-2**, and a third die **114-3** with the second pitch coupled to the second surface **170-2** of the die-level interposer **104** by a third HB region **130-3**. As shown in FIG. 2C, the die-level interposer **104** may include first and second standoff layers **101-1**, **101-2** to provide distance (e.g., z-height) between the first die **114-1** and the second and third dies **114-2**, **114-3** and may include TSVs **113** to provide vertical pitch translation between the first and second standoff layers **101-1**, **101-2**, where a same number of TSVs in the first standoff layer **101-1** are coupled to a same number of TSVs in the second standoff layer **101-2** (e.g., one TSV in the first standoff layer **101-1** to one TSV in the second standoff layer **101-2**). A standoff layer may have any suitable dimensions, as described above with reference to FIG. 1. Although FIG. 2A-2C show the first and second standoff layers **101-1**, **101-2** as having a same thickness, a standoff layer **101** may have any suitable thickness and different standoff layers **101** may have different thicknesses. For example, a first standoff layer **101-1** may have a thickness between 5 microns and 400 microns and a second standoff layer **101-2** may have a thickness between 5 microns and 400 microns.

[0045] FIG. 3 is a side, cross-sectional view of an example microelectronic assembly, in accordance with various embodiments. FIG. 3 is an example microelectronic assembly **100** including a die-level interposer **104** formed on a first die **114-1**, a second die **114-2** with a second pitch coupled to the second surface **170-2** of the die-level interposer **104** by a HB region **130-2**, and a third die **114-3** with a third pitch different from the second pitch coupled to the second surface **170-2** of the die-level interposer **104** by a HB region **130-3**. The die-level interposer **104** may be fabricated by depositing a dielectric material **105** on a surface (e.g., non-active or backside) of the

first die **114-1** and patterning conductive pathways **111** through the dielectric material (e.g., using BEOL processes, as known in the art, and, in such embodiments, a thickness of the die-level interposer **104** may be between 5 microns and 30 microns). In some embodiments, the die-level interposer **104** may be fabricated using organic material (e.g., using RDL processes, as known in the art, and in such embodiments, a thickness of the die-level interposer **104** may be between 20 microns and 100 microns). The HB interface **180** (not labeled) may be formed on the top surface of the dielectric material **105** subsequent to patterning the conductive pathways **111**.

[0046] FIGS. **4A** and **4B** are side, cross-sectional views of example microelectronic assemblies, in accordance with various embodiments. FIG. **4A** is an example microelectronic assembly **100** including a die-level interposer **104** having a first standoff layer **101-1**, a routing layer **148**, and a second standoff layer **101-2**. As shown in FIG. **4A**, the die-level interposer **104** may include a first standoff layer **101-1** at a first surface **170-1**, a routing layer **148** on the first standoff layer **101-1** and coupled to the first standoff layer **101-1** by a HB region **130-6**, and a second standoff layer **101-2** on the routing layer **148** (e.g., at the second surface **170-2**). As shown in FIG. **4A**, the microelectronic assembly **100** may further include a first die **114-1** with a first pitch coupled to a first surface **170-1** of the die-level interposer **104** by a first hybrid bonding (HB) region **130-1**, a second die **114-2** with a second pitch different from the first pitch coupled to the second surface **170-2** of the die-level interposer **104** by a second HB region **130-2**, and a third die **114-3** with a third pitch different from the first and second pitches coupled to the second surface **170-2** of the die-level interposer **104** by a third HB region **130-3**. The first and second standoff layers **101-1**, **101-2** may include TSVs **113** and the routing layer **148** may include conductive pathways **111** that provide pitch translation between the first pitch of the first die **114-1** and the second and third pitches of the second and third dies **114-2**, **114-3**, respectively.

[0047] FIG. **4B** is an example microelectronic assembly **100** including a die-level interposer **104** having a first standoff layer **101-1**, a second standoff layer **101-2**, a routing layer **148**, and a third standoff layer **101-3**. As shown in FIG. **4B**, the die-level interposer **104** may include a first standoff layer **101-1** at a first surface **170-1**, a second standoff layer **101-2** on the first standoff layer **101-1** and coupled to the first standoff layer **101-1** by a HB region **130-5**, a routing layer **148** on the second standoff layer **101-2** and coupled to the second standoff layer **101-2** by a HB region **130-6**, and a third standoff layer **101-3** on the routing layer **148** (e.g., at the second surface **170-2**). As shown in FIG. **4B**, the microelectronic assembly **100** may further include a first die **114-1** with a first pitch coupled to a first surface **170-1** of the die-level interposer **104** by a first hybrid bonding (HB) region **130-1**, a second die **114-2** with a second pitch different from the first pitch coupled to the second surface **170-2** of the die-level interposer **104** by a second HB region **130-2**, and a third die **114-3** with a third pitch different from the first and second pitches coupled to the second surface **170-2** of the die-level interposer **104** by a third HB region **130-3**. The first, second, and third standoff layers **101-1**, **101-2**, **101-3** may include TSVs **113** and the routing layer **148** may include conductive pathways **111** that provide pitch translation between the first pitch of the first die **114-1** and the second and third pitches of the second and third dies **114-2**, **114-3**, respectively.

[0048] Any suitable techniques may be used to manufacture the microelectronic assemblies disclosed herein. FIGS. **5A-5F** are side, cross-sectional views of various stages in an example process for manufacturing a microelectronic assembly, in accordance with various embodiments. Although the operations discussed below with reference to FIGS. **5A-5F** (and others of the accompanying drawings representing manufacturing processes) are illustrated in a particular order, these operations may be performed in any suitable order. FIG. **5A** illustrates a first standoff layer **101-1** (e.g., the first standoff layer **101-1** of the die-level interposer **104** of FIG. **2A**) having TSVs **113** and HB interfaces **180-5A**, **180-1B**. The first standoff layer **101-1** may include a wafer having TSVs **113** and top dielectric and metallization layers. In some embodiments, additional processing operations may be performed on the first standoff layer **101-1**, for example, wafer thinning, top surface TSV **113** reveal, TSV pad creation, and forming HB interfaces **180-5A**, **180-1B**. In some

embodiments, the first standoff layer **101-1** may be mounted on a carrier prior to performing additional processing operations. The carrier may include any suitable material, and in some embodiments, may include a semiconductor wafer (e.g., a silicon wafer) or glass (e.g., a glass panel).

[0049] FIG. 5B illustrates an assembly subsequent to hybrid bonding the first standoff layer **101-1** to a first die **114-1** having HB interfaces **180-1A** (not labeled), **180-4B**. In particular, HB interface **180-1B** may be brought into contact with HB interface **180** of the first die **114-1** (not labeled), and heat and/or pressure to be applied to bond the contacting HB interfaces **180** to form HB region **130-1**. In some embodiments, the first die **114-1** may be mounted on a carrier prior to forming HB region **130-1**.

[0050] FIG. 5C illustrates a second standoff layer **101-2** (e.g., the second standoff layer **101-2** of the die-level interposer **104** of FIG. 2A) having TSVs **113** and HB interfaces **180-5B**, **180-2A**, **180-3A**. The second standoff layer **101-2** may include a wafer having TSVs **113** and/or top dielectric and metallization layers. In some embodiments, additional processing operations may be performed on the second standoff layer **101-2**, for example, wafer thinning, top surface TSV **113** reveal, TSV pad creation, and forming HB interfaces **180-5B**, **180-2A**, **180-3A**. In some embodiments, the second standoff layer **101-2** may be mounted on a carrier prior to performing additional processing operations.

[0051] FIG. 5D illustrates an assembly subsequent to hybrid bonding the second standoff layer **101-2** on the first standoff layer **101-1** of the assembly of FIG. 5B. In particular, HB interface **180-5B** (not labeled) of the second standoff layer **101-2** may be brought into contact with HB interface **180-5A** (not labeled) of the first standoff layer **101-1**, and heat and/or pressure to be applied to bond the contacting HB interfaces **180** to form HB region **130-5**. The hybrid bonded first and second standoff layers **101-1**, **101-2** may form a die-level interposer **104**.

[0052] FIG. 5E illustrates an assembly subsequent to hybrid bonding the second and third dies **114-2**, **114-3** to the second standoff layer **101-2** of the assembly of FIG. 5D. In particular, HB interface **180-2B** (not labeled) of the second die **114-2** may be brought into contact with HB interface **180-2A** (not labeled) of the second standoff layer **101-2** and HB interface **180-3B** (not labeled) of the third die **114-3** may be brought into contact with HB interface **180-3A** (not labeled) of the second standoff layer **101-2**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB regions **130-2** and **130-3**, respectively.

[0053] FIG. 5F illustrates an assembly subsequent to hybrid bonding the first die **114-1** to a substrate **150**, providing an insulating material **126** around the second and third dies **114-2**, **114-3**, and providing solder **120** on the conductive contacts **118**. In particular, HB interface **180-4B** (not labeled) of the first die **114-1** may be brought into contact with the HB interface **180** (not labeled) of the substrate **150**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB region **130-4**. The insulating material **126** may be formed using any suitable process, including lamination, or slit coating and curing. In some embodiments, the insulating material **126** may extend above and remain above the second and third dies **114-2**, **114-3**, while in other embodiments, the insulating material **126** may be polished back to expose the top surfaces of the second and third dies **114-2**, **114-3**, as shown. In some embodiments, the insulating material **126** may be provided prior to hybrid bonding the first die **114-1** on the substrate **150**. In some embodiments, the insulating material **126** may be provided subsequent to hybrid bonding the first die **114-1** on the substrate **150**. In some embodiments, the substrate **150** may be mounted on a carrier prior to hybrid bonding the first die **114-1** and the carrier may be removed prior to providing solder **120** on the conductive contacts **118**. The assembly of FIG. 5F may itself be a microelectronic assembly **100**, as shown. Further manufacturing operations may be performed on the microelectronic assembly **100** of FIG. 5F to form other microelectronic assemblies **100**; for example, the solder **120** may be used to couple the microelectronic assembly **100** of FIG. 5F to a support component **182**, and a TIM **154** and heat transfer structure **156** may be provided on the top

surface of the microelectronic assembly **100** of FIG. 5F, similar to the microelectronic assembly **100** of FIG. 1A.

[0054] Microelectronic assemblies **100** including multiple tiers of standoff layers **101** may be formed in a manner discussed above with reference to FIGS. 5A-5F, with the additional tiers of standoff layers **101** coupled to the preceding assemblies prior to deposition of the second and third dies **114-2**, **114-3** and the insulating material **126**.

[0055] FIGS. 6A-6E are side, cross-sectional views of various stages in an example process for manufacturing the microelectronic assembly of FIG. 1A, in accordance with various embodiments. FIG. 6A illustrates a standoff layer **101** (e.g., the standoff layer **101** of the die-level interposer **104** of FIG. 1A) having TSVs **113** and HB interfaces **180-2A**, **180-3A**. The standoff layer **101** may include a wafer having TSVs **113** and top dielectric and metallization layers. In some embodiments, additional processing operations may be performed on the standoff layer **101**, for example, wafer thinning, top surface TSV reveal, TSV pad creation, and forming HB interfaces **180-2A**, **180-3A**. In some embodiments, the standoff layer **101** may be mounted on a carrier prior to performing additional processing operations.

[0056] FIG. 6B illustrates an assembly subsequent to forming a routing layer **148** on the standoff layer **101** and forming HB interface **180-1B** on the routing layer **148**. The routing layer **148** may be formed on the standoff layer **101** using BEOL processing, as known in the art. The standoff layer **101** and routing layer **148** may form a die-level interposer **104**.

[0057] FIG. 6C illustrates an assembly subsequent to hybrid bonding the routing layer **148** of the die-level interposer **104** to a first die **114-1** having HB interfaces **180-1A** (not labeled), **180-4B**. In particular, HB interface **180-1B** of the assembly of FIG. 6B may be brought into contact with HB interface **180** of the first die **114-1** (not labeled), and heat and/or pressure to be applied to bond the contacting HB interfaces **180** to form HB region **130-1**. In some embodiments, the first die **114-1** may be mounted on a carrier prior to forming HB region **130-1**.

[0058] FIG. 6D illustrates an assembly subsequent to hybrid bonding the second and third dies **114-2**, **114-3** to the standoff layer **101** of the assembly of FIG. 6C. In particular, HB interface **180-2B** (not labeled) of the second die **114-2** may be brought into contact with HB interface **180-2A** (not labeled) of the standoff layer **101** and HB interface **180-3B** (not labeled) of the third die **114-3** may be brought into contact with HB interface **180-3A** (not labeled) of the standoff layer **101**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB regions **130-2** and **130-3**, respectively.

[0059] FIG. 6E illustrates an assembly subsequent to hybrid bonding the first die **114-1** to a substrate **150**, providing an insulating material **126** around the second and third dies **114-2**, **114-3**, and providing solder **120** on the conductive contacts **118**. In particular, HB interface **180-4B** (not labeled) of the first die **114-1** may be brought into contact with the HB interface **180** (not labeled) of the substrate **150**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB region **130-4**. The insulating material **126** may be formed using any suitable process, including lamination, or slit coating and curing. In some embodiments, the insulating material **126** may extend above and remain above the second and third dies **114-2**, **114-3**, while in other embodiments, the insulating material **126** may be polished back to expose the top surfaces of the second and third dies **114-2**, **114-3**, as shown. In some embodiments, the insulating material **126** may be provided prior to hybrid bonding the first die **114-1** on the substrate **150**. In some embodiments, the insulating material **126** may be provided subsequent to hybrid bonding the first die **114-1** on the substrate **150**. In some embodiments, the substrate **150** may be mounted on a carrier prior to hybrid bonding the first die **114-1** and the carrier may be removed prior to providing solder **120** on the conductive contacts **118**. The assembly of FIG. 6E may itself be a microelectronic assembly **100**, as shown. Further manufacturing operations may be performed on the microelectronic assembly **100** of FIG. 6E to form other microelectronic assemblies **100**; for example, the solder **120** may be used to couple the microelectronic assembly **100** of FIG. 6E to a

support component **182**, and a TIM **154** and heat transfer structure **156** may be provided on the top surface of the microelectronic assembly **100** of FIG. **6E**, similar to the microelectronic assembly **100** of FIG. **1A**.

[0060] Microelectronic assemblies **100** including multiple tiers of standoff layers **101** and routing layers **148** may be formed in a manner discussed above with reference to FIGS. **6A-6E**, with the additional tiers of standoff layers **101** and routing layers **148** coupled to the preceding assemblies prior to deposition of the second and third dies **114-2**, **114-3** and the insulating material **126**.

[0061] FIGS. **7A-7E** are side, cross-sectional views of various stages in an example process for manufacturing a microelectronic assembly, in accordance with various embodiments. FIG. **7A** illustrates a first die **114-1** having HB interface **180-1B** at a bottom surface and conductive contacts **710** at a top surface. In some embodiments, the first die **114-1** may be mounted on a carrier prior to performing additional processing operations.

[0062] FIG. **7B** illustrates an assembly subsequent to depositing a dielectric material **105** on the top surface of the first die **114-1** and forming conductive pathways **111** in the dielectric material **105**. The dielectric material **105** may be deposited and the conductive pathways **111** may be formed using any suitable technique, for example, using a BEOL process or an RDL process. Additional dielectric and conductive layers may be formed by repeating the BEOL process or RDL process (e.g., as shown in FIG. **7C**), as known in the art.

[0063] FIG. **7C** illustrates an assembly subsequent to depositing additional dielectric material **105**, forming additional conductive pathways **111** in the dielectric material **105**, and forming HB interfaces **180-2A**, **180-3A** on the top surface of the dielectric material **105**. The dielectric material **105** and conductive pathways **111** may form a die-level interposer **104**.

[0064] FIG. **7D** illustrates an assembly subsequent to hybrid bonding the second and third dies **114-2**, **114-3** to the top surface of the assembly of FIG. **7C**. In particular, HB interface **180-2B** (not labeled) of the second die **114-2** may be brought into contact with HB interface **180-2A** (not labeled) of the die-level interposer **104** and HB interface **180-3B** (not labeled) of the third die **114-3** may be brought into contact with HB interface **180-3A** (not labeled) of the die-level interposer **104**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB regions **130-2** and **130-3**, respectively.

[0065] FIG. **7E** illustrates an assembly subsequent to hybrid bonding the first die **114-1** to a substrate **150**, providing an insulating material **126** around the second and third dies **114-2**, **114-3**, and providing solder **120** on the conductive contacts **118**. In particular, HB interface **180-4B** (not labeled) of the first die **114-1** may be brought into contact with the HB interface **180** (not labeled) of the substrate **150**, and heat and/or pressure may be applied to bond the contacting HB interfaces **180** to form HB region **130-4**. The insulating material **126** may be formed using any suitable process, including lamination, or slit coating and curing. In some embodiments, the insulating material **126** may extend above and remain above the second and third dies **114-2**, **114-3**, while in other embodiments, the insulating material **126** may be polished back to expose the top surfaces of the second and third dies **114-2**, **114-3**, as shown. In some embodiments, the insulating material **126** may be provided prior to hybrid bonding the first die **114-1** on the substrate **150**. In some embodiments, the insulating material **126** may be provided subsequent to hybrid bonding the first die **114-1** on the substrate **150**. In some embodiments, the substrate **150** may be mounted on a carrier prior to hybrid bonding the first die **114-1** and the carrier may be removed prior to providing solder **120** on the conductive contacts **118**. The assembly of FIG. **7E** may itself be a microelectronic assembly **100**, as shown. Further manufacturing operations may be performed on the microelectronic assembly **100** of FIG. **7E** to form other microelectronic assemblies **100**; for example, the solder **120** may be used to couple the microelectronic assembly **100** of FIG. **7E** to a support component **182**, and a TIM **154** and heat transfer structure **156** may be provided on the top surface of the microelectronic assembly **100** of FIG. **7E**, similar to the microelectronic assembly **100** of FIG. **1A**.

[0066] The microelectronic assemblies **100** disclosed herein may be included in any suitable electronic component. FIGS. **8-11** illustrate various examples of apparatuses that may include, or be included in, as suitable, any of the microelectronic assemblies **100** disclosed herein.

[0067] FIG. **8** is a top view of a wafer **1500** and dies **1502** that may be included in any of the microelectronic assemblies **100** disclosed herein. For example, a die **1502** may serve as a die **114**, or may be included in a die **114**. The wafer **1500** may be composed of semiconductor material and may include one or more dies **1502** having IC structures formed on a surface of the wafer **1500**. Each of the dies **1502** may be a repeating unit of a semiconductor product that includes any suitable IC. After the fabrication of the semiconductor product is complete, the wafer **1500** may undergo a singulation process in which the dies **1502** are separated from one another to provide discrete “chips” of the semiconductor product. The die **1502** may include one or more transistors (e.g., some of the transistors **1640** of FIG. **9**, discussed below) and/or supporting circuitry to route electrical signals to the transistors, as well as any other IC components. In some embodiments, the wafer **1500** or the die **1502** may include a memory device (e.g., a random access memory (RAM) device, such as a static RAM (SRAM) device, a magnetic RAM (MRAM) device, a resistive RAM (RRAM) device, a conductive-bridging RAM (CBRAM) device, etc.), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **1502**. For example, a memory array formed by multiple memory devices may be formed on a same die **1502** as a processing device (e.g., the processing device **1802** of FIG. **11**) or other logic that is configured to store information in the memory devices or execute instructions stored in the memory array.

[0068] FIG. **9** is a side, cross-sectional view of an IC device **1600** that may be included in any of the die **114** disclosed herein. For example, an IC device **1600** (e.g., as part of a die **1502**, as discussed above with reference to FIG. **8**) may serve as a die **114**, or may be included in a die **114**. One or more of the IC devices **1600** may be included in one or more dies **1502** (FIG. **8**). The IC device **1600** may be formed on a substrate **1602** (e.g., the wafer **1500** of FIG. **8**) and may be included in a die (e.g., the die **1502** of FIG. **8**). The substrate **1602** may be a semiconductor substrate composed of semiconductor material systems including, for example, n-type or p-type materials systems (or a combination of both). The substrate **1602** may include, for example, a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In some embodiments, the substrate **1602** may be formed using alternative materials, which may or may not be combined with silicon, that include but are not limited to germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the substrate **1602**. Although a few examples of materials from which the substrate **1602** may be formed are described here, any material that may serve as a foundation for an IC device **1600** may be used. The substrate **1602** may be part of a singulated die (e.g., the dies **1502** of FIG. **8**) or a wafer (e.g., the wafer **1500** of FIG. **8**).

[0069] The IC device **1600** may include one or more device layers **1604** disposed on the substrate **1602**. The device layer **1604** may include features of one or more transistors **1640** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the substrate **1602**. The device layer **1604** may include, for example, one or more source and/or drain (S/D) regions **1620**, a gate **1622** to control current flow in the transistors **1640** between the S/D regions **1620**, and one or more S/D contacts **1624** to route electrical signals to/from the S/D regions **1620**. The transistors **1640** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **1640** are not limited to the type and configuration depicted in FIG. **9** and may include a wide variety of other types and configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Planar transistors may include bipolar junction transistors (BJT), heterojunction bipolar transistors (HBT), or high-electron-mobility transistors (HEMT). Non-planar transistors may include FinFET transistors, such

as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors.

[0070] Each transistor **1640** may include a gate **1622** formed of at least two layers, a gate dielectric and a gate electrode. The gate dielectric may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material. The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process may be carried out on the gate dielectric to improve its quality when a high-k material is used.

[0071] The gate electrode may be formed on the gate dielectric and may include at least one p-type work function metal or n-type work function metal, depending on whether the transistor **1640** is to be a p-type metal oxide semiconductor (PMOS) or an n-type metal oxide semiconductor (NMOS) transistor. In some implementations, the gate electrode may consist of a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as a barrier layer. For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and any of the metals discussed below with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide), and any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning).

[0072] In some embodiments, when viewed as a cross-section of the transistor **1640** along the source-channel-drain direction, the gate electrode may consist of a U-shaped structure that includes a bottom portion substantially parallel to the surface of the substrate and two sidewall portions that are substantially perpendicular to the top surface of the substrate. In other embodiments, at least one of the metal layers that form the gate electrode may simply be a planar layer that is substantially parallel to the top surface of the substrate and does not include sidewall portions substantially perpendicular to the top surface of the substrate. In other embodiments, the gate electrode may consist of a combination of U-shaped structures and planar, non-U-shaped structures. For example, the gate electrode may consist of one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

[0073] In some embodiments, a pair of sidewall spacers may be formed on opposing sides of the gate stack to bracket the gate stack. The sidewall spacers may be formed from materials such as silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, and silicon oxynitride. Processes for forming sidewall spacers are well known in the art and generally include deposition and etching process steps. In some embodiments, a plurality of spacer pairs may be used; for instance, two pairs, three pairs, or four pairs of sidewall spacers may be formed on opposing sides of the gate stack.

[0074] The S/D regions **1620** may be formed within the substrate **1602** adjacent to the gate **1622** of each transistor **1640**. The S/D regions **1620** may be formed using an implantation/diffusion process or an etching/deposition process, for example. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the substrate **1602** to form the S/D regions **1620**. An annealing process that activates the dopants and causes them to diffuse farther into the substrate **1602** may follow the ion-implantation process. In the latter process, the

substrate **1602** may first be etched to form recesses at the locations of the S/D regions **1620**. An epitaxial deposition process may then be carried out to fill the recesses with material that is used to fabricate the S/D regions **1620**. In some implementations, the S/D regions **1620** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some embodiments, the S/D regions **1620** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further embodiments, one or more layers of metal and/or metal alloys may be used to form the S/D regions **1620**.

[0075] Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., the transistors **1640**) of the device layer **1604** through one or more interconnect layers disposed on the device layer **1604** (illustrated in FIG. **9** as interconnect layers **1606-1610**). For example, electrically conductive features of the device layer **1604** (e.g., the gate **1622** and the S/D contacts **1624**) may be electrically coupled with the interconnect structures **1628** of the interconnect layers **1606-1610**. The one or more interconnect layers **1606-1610** may form a metallization stack (also referred to as an “ILD stack”) **1619** of the IC device **1600**.

[0076] The interconnect structures **1628** may be arranged within the interconnect layers **1606-1610** to route electrical signals according to a wide variety of designs (in particular, the arrangement is not limited to the particular configuration of interconnect structures **1628** depicted in FIG. **9**). Although a particular number of interconnect layers **1606-1610** is depicted in FIG. **9**, embodiments of the present disclosure include IC devices having more or fewer interconnect layers than depicted.

[0077] In some embodiments, the interconnect structures **1628** may include lines **1628a** and/or vias **1628b** filled with an electrically conductive material such as a metal. The lines **1628a** may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the substrate **1602** upon which the device layer **1604** is formed. For example, the lines **1628a** may route electrical signals in a direction in and out of the page from the perspective of FIG. **9**. The vias **1628b** may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the substrate **1602** upon which the device layer **1604** is formed. In some embodiments, the vias **1628b** may electrically couple lines **1628a** of different interconnect layers **1606-1610** together.

[0078] The interconnect layers **1606-1610** may include a dielectric material **1626** disposed between the interconnect structures **1628**, as shown in FIG. **9**. In some embodiments, the dielectric material **1626** disposed between the interconnect structures **1628** in different ones of the interconnect layers **1606-1610** may have different compositions; in other embodiments, the composition of the dielectric material **1626** between different interconnect layers **1606-1610** may be the same.

[0079] A first interconnect layer **1606** may be formed above the device layer **1604**. In some embodiments, the first interconnect layer **1606** may include lines **1628a** and/or vias **1628b**, as shown. The lines **1628a** of the first interconnect layer **1606** may be coupled with contacts (e.g., the S/D contacts **1624**) of the device layer **1604**.

[0080] A second interconnect layer **1608** may be formed above the first interconnect layer **1606**. In some embodiments, the second interconnect layer **1608** may include vias **1628b** to couple the lines **1628a** of the second interconnect layer **1608** with the lines **1628a** of the first interconnect layer **1606**. Although the lines **1628a** and the vias **1628b** are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer **1608**) for the sake of clarity, the lines **1628a** and the vias **1628b** may be structurally and/or materially contiguous (e.g., simultaneously filled during a dual-damascene process) in some embodiments.

[0081] A third interconnect layer **1610** (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **1608** according to similar techniques and configurations described in connection with the second interconnect layer **1608** or the first

interconnect layer **1606**. In some embodiments, the interconnect layers that are “higher up” in the metallization stack **1619** in the IC device **1600** (i.e., farther away from the device layer **1604**) may be thicker.

[0082] The IC device **1600** may include a solder resist material **1634** (e.g., polyimide or similar material) and one or more conductive contacts **1636** formed on the interconnect layers **1606-1610**. In FIG. **9**, the conductive contacts **1636** are illustrated as taking the form of bond pads. The conductive contacts **1636** may be electrically coupled with the interconnect structures **1628** and configured to route the electrical signals of the transistor(s) **1640** to other external devices. For example, solder bonds may be formed on the one or more conductive contacts **1636** to mechanically and/or electrically couple a chip including the IC device **1600** with another component (e.g., a circuit board). The IC device **1600** may include additional or alternate structures to route the electrical signals from the interconnect layers **1606-1610**; for example, the conductive contacts **1636** may include other analogous features (e.g., posts) that route the electrical signals to external components.

[0083] FIG. **10** is a side, cross-sectional view of an IC device assembly **1700** that may include any of the die **114** and/or microelectronic assemblies **100** disclosed herein. The IC device assembly **1700** includes a number of components disposed on a circuit board **1702** (which may be, e.g., a motherboard). The IC device assembly **1700** includes components disposed on a first face **1740** of the circuit board **1702** and an opposing second face **1742** of the circuit board **1702**; generally, components may be disposed on one or both faces **1740** and **1742**. Any of the IC packages discussed below with reference to the IC device assembly **1700** may include any of the embodiments of the microelectronic assemblies **100** disclosed herein (e.g., may include multiple die **114** coupled together by hybrid bonding).

[0084] In some embodiments, the circuit board **1702** may be a PCB including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1702**. In other embodiments, the circuit board **1702** may be a non-PCB substrate.

[0085] The IC device assembly **1700** illustrated in FIG. **10** includes a package-on-interposer structure **1736** coupled to the first face **1740** of the circuit board **1702** by coupling components **1716**. The coupling components **1716** may electrically and mechanically couple the package-on-interposer structure **1736** to the circuit board **1702**, and may include solder balls (as shown in FIG. **10**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

[0086] The package-on-interposer structure **1736** may include an IC package **1720** coupled to a package interposer **1704** by coupling components **1718**. The coupling components **1718** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1716**. Although a single IC package **1720** is shown in FIG. **10**, multiple IC packages may be coupled to the package interposer **1704**; indeed, additional interposers may be coupled to the package interposer **1704**. The package interposer **1704** may provide an intervening substrate used to bridge the circuit board **1702** and the IC package **1720**. The IC package **1720** may be or include, for example, a die (the die **1502** of FIG. **8**), an IC device (e.g., the IC device **1600** of FIG. **9**), or any other suitable component. Generally, the package interposer **1704** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the package interposer **1704** may couple the IC package **1720** (e.g., a die) to a set of BGA conductive contacts of the coupling components **1716** for coupling to the circuit board **1702**. In the embodiment illustrated in FIG. **10**, the IC package **1720** and the circuit board **1702** are attached to opposing sides of the package interposer **1704**; in other embodiments, the IC package **1720** and the circuit board **1702** may be attached to a same side of the package interposer **1704**. In some

embodiments, three or more components may be interconnected by way of the package interposer **1704**.

[0087] In some embodiments, the package interposer **1704** may be formed as a PCB, including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. In some embodiments, the package interposer **1704** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, an epoxy resin with inorganic fillers, a ceramic material, or a polymer material such as polyimide. In some embodiments, the package interposer **1704** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The package interposer **1704** may include metal lines **1710** and vias **1708**, including but not limited to TSVs **1706**. The package interposer **1704** may further include embedded devices **1714**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) devices, and memory devices. More complex devices such as radio frequency devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the package interposer **1704**. The package-on-interposer structure **1736** may take the form of any of the package-on-interposer structures known in the art.

[0088] The IC device assembly **1700** may include an IC package **1724** coupled to the first face **1740** of the circuit board **1702** by coupling components **1722**. The coupling components **1722** may take the form of any of the embodiments discussed above with reference to the coupling components **1716**, and the IC package **1724** may take the form of any of the embodiments discussed above with reference to the IC package **1720**.

[0089] The IC device assembly **1700** illustrated in FIG. **10** includes a package-on-package structure **1734** coupled to the second face **1742** of the circuit board **1702** by coupling components **1728**. The package-on-package structure **1734** may include an IC package **1726** and an IC package **1732** coupled together by coupling components **1730** such that the IC package **1726** is disposed between the circuit board **1702** and the IC package **1732**. The coupling components **1728** and **1730** may take the form of any of the embodiments of the coupling components **1716** discussed above, and the IC packages **1726** and **1732** may take the form of any of the embodiments of the IC package **1720** discussed above. The package-on-package structure **1734** may be configured in accordance with any of the package-on-package structures known in the art.

[0090] FIG. **11** is a block diagram of an example electrical device **1800** that may include any of the die **114** and/or microelectronic assemblies **100** disclosed herein. For example, any suitable ones of the components of the electrical device **1800** may include one or more of the IC device assemblies **1700**, IC devices **1600**, or dies **1502** disclosed herein. A number of components are illustrated in FIG. **11** as included in the electrical device **1800**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the electrical device **1800** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single system-on-a-chip (SoC) die.

[0091] Additionally, in various embodiments, the electrical device **1800** may not include one or more of the components illustrated in FIG. **11**, but the electrical device **1800** may include interface circuitry for coupling to the one or more components. For example, the electrical device **1800** may not include a display device **1806**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **1806** may be coupled. In another set of examples, the electrical device **1800** may not include an audio input device **1824** or an audio output device **1808**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1824** or audio output device **1808** may be coupled.

[0092] The electrical device **1800** may include a processing device **1802** (e.g., one or more processing devices). As used herein, the term “processing device” or “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processing device **1802** may include one or more digital signal processors (DSPs), application-specific integrated circuits (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The electrical device **1800** may include a memory **1804**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random access memory (DRAM)), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some embodiments, the memory **1804** may include memory that shares a die with the processing device **1802**. This memory may be used as cache memory and may include embedded dynamic random access memory (eDRAM) or spin transfer torque magnetic random access memory (STT-MRAM).

[0093] In some embodiments, the electrical device **1800** may include a communication chip **1812** (e.g., one or more communication chips). For example, the communication chip **1812** may be configured for managing wireless communications for the transfer of data to and from the electrical device **1800**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

[0094] The communication chip **1812** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra mobile broadband (UMB) project (also referred to as “3GPP2”), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1812** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1812** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1812** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1812** may operate in accordance with other wireless protocols in other embodiments. The electrical device **1800** may include an antenna **1822** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

[0095] In some embodiments, the communication chip **1812** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1812** may include multiple communication chips. For instance, a first communication chip **1812** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1812** may be dedicated to longer-range wireless communications such as global positioning system (GPS),

EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication chip **1812** may be dedicated to wireless communications, and a second communication chip **1812** may be dedicated to wired communications.

[0096] The electrical device **1800** may include battery/power circuitry **1814**. The battery/power circuitry **1814** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the electrical device **1800** to an energy source separate from the electrical device **1800** (e.g., AC line power).

[0097] The electrical device **1800** may include a display device **1806** (or corresponding interface circuitry, as discussed above). The display device **1806** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display.

[0098] The electrical device **1800** may include an audio output device **1808** (or corresponding interface circuitry, as discussed above). The audio output device **1808** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds.

[0099] The electrical device **1800** may include an audio input device **1824** (or corresponding interface circuitry, as discussed above). The audio input device **1824** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

[0100] The electrical device **1800** may include a GPS device **1818** (or corresponding interface circuitry, as discussed above). The GPS device **1818** may be in communication with a satellite-based system and may receive a location of the electrical device **1800**, as known in the art.

[0101] The electrical device **1800** may include an other output device **1810** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1810** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.

[0102] The electrical device **1800** may include an other input device **1820** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1820** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

[0103] The electrical device **1800** may have any desired form factor, such as a handheld or mobile electrical device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultra mobile personal computer, etc.), a desktop electrical device, a server device or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable electrical device. In some embodiments, the electrical device **1800** may be any other electronic device that processes data.

[0104] The following paragraphs provide various examples of the embodiments disclosed herein.

[0105] Example 1 is a microelectronic assembly, comprising: a die-level interposer having a first surface and an opposing second surface; a first die coupled to the first surface of the die-level interposer by a first hybrid bonding region having a first pitch; and a second die coupled to the second surface of the die-level interposer by a second hybrid bonding region having a second pitch different from the first pitch.

[0106] Example 2 may include the subject matter of Example 1, and may further specify that the first pitch has a pitch between 1 micron and 10 microns.

[0107] Example 3 may include the subject matter of Examples 1 or 2, and may further specify that the second pitch has a pitch between 1 microns and 10 microns.

[0108] Example 4 may include the subject matter of Example 2, and may further specify that the second pitch has a pitch between 10 microns and 200 microns.

[0109] Example 5 may include the subject matter of Example 1, and may further specify that a thickness of the die-level interposer is between 5 microns and 1000 microns.

[0110] Example 6 may include the subject matter of Example 1, and may further specify that the die-level interposer includes a routing layer; and a substrate layer on the routing layer.

[0111] Example 6B may include the subject matter of Example 1, and may further specify that the die-level interposer further includes a routing layer, wherein the first die is coupled to the routing layer by the first hybrid bonding region; and a substrate layer on the routing layer, wherein the second die is coupled to the substrate layer by the second hybrid bonding region.

[0112] Example 7 may include the subject matter of Example 6, and may further specify that a material of the routing layer includes one or more of silicon and oxygen, an organic dielectric, an inorganic dielectric, a polyimide material, an organic polymeric dielectric, a photo-imageable dielectric, a benzocyclobutene-based polymer, or a ceramic.

[0113] Example 8 may include the subject matter of Example 6, and may further specify that a material of the substrate layer includes one or more of silicon, silicon and oxygen, silicon and carbon, oxynitride, a polyimide material, fused silica, glass, a glass-reinforced epoxy matrix material, a carbon-doped dielectric, a fluorine-doped dielectric, a porous dielectric, an organic polymeric dielectric, a photo-imageable dielectric, a benzocyclobutene-based polymer, germanium, a III-V material, an organic dielectric, and an organic polymer matrix.

[0114] Example 9 may include the subject matter of Example 6, and may further specify that a thickness of the routing layer is between Example 0.05 microns and 5 microns.

[0115] Example 10 may include the subject matter of Example 6, and may further specify that a thickness of the substrate layer is between 5 microns and 400 microns.

[0116] Example 11 is a microelectronic assembly, including a first die; a second die; a third die; and a die-level interposer, having a first surface and an opposing second surface, coupled to the first die at the first surface by a first hybrid bonding region having a first pitch, coupled to the second die at the second surface by a second hybrid bonding region having a second pitch different from the first pitch, and coupled to the third die at the second surface by a third hybrid bonding region having a third pitch different from the first pitch.

[0117] Example 12 may include the subject matter of Example 11, and may further specify that the die-level interposer further includes a first substrate layer having a first through-substrate via (TSV); and a second substrate layer, having a second TSV, on the first substrate layer and coupled to the first substrate layer by a fourth hybrid bonding region having a fourth pitch different from the first pitch, and wherein the first TSV is coupled to the second TSV by the fourth hybrid bonding region.

[0118] Example 13 may include the subject matter of Example 12, and may further specify that the fourth pitch is a different pitch from the second pitch and the third pitch.

[0119] Example 14 may include the subject matter of Example 12, and may further specify that the fourth pitch is a same pitch as the second or third pitch.

[0120] Example 15 may include the subject matter of Example 12, and may further specify that a material of the first and second substrate layers includes one or more of silicon, silicon and oxygen, silicon and carbon, oxynitride, a polyimide material, fused silica, glass, a glass-reinforced epoxy matrix material, a carbon-doped dielectric, a fluorine-doped dielectric, a porous dielectric, an organic polymeric dielectric, a photo-imageable dielectric, a benzocyclobutene-based polymer, germanium, a III-V material, an organic dielectric, and an organic polymer matrix.

[0121] Example 16 may include the subject matter of Example 12, and may further specify that a thickness of the first substrate layer is between 5 microns and 400 microns and a thickness of the second substrate layer is between 5 microns and 400 microns.

[0122] Example 17 may include the subject matter of Example 11, and may further specify that the first die has a first surface and an opposing second surface and the die-level interposer is coupled to the second surface of the first die, and may further include a package substrate coupled to the first

surface of the first die by a fourth hybrid bonding region.

[0123] Example 18 is a method of manufacturing a microelectronic assembly, including coupling a first surface of a die-level interposer, having the first surface and an opposing second surface, to a first die by a first hybrid bonding region with a first pitch; and coupling the second surface of the die-level interposer to a second die by a second hybrid bonding region having a second pitch different from the first pitch.

[0124] Example 19 may include the subject matter of Example 18, and may further specify that the die-level interposer includes a first substrate layer and a second substrate layer on the first substrate layer, and wherein the first substrate layer is coupled to the second substrate layer by a third hybrid bonding region.

[0125] Example 20 may include the subject matter of Example 18, and may further specify that the die-level interposer includes a routing layer and a substrate layer on the routing layer.

[0126] Example 21 may include the subject matter of Example 18, and may further specify that the first pitch has a pitch between 1 micron and 10 microns and the second pitch has a pitch between 1 micron and 10 microns.

[0127] Example 22 may include the subject matter of Example 18, and may further specify that the first pitch has a pitch between 1 micron and 10 microns and the second pitch has a pitch between 10 microns and 200 microns.

Claims

1. An assembly, comprising: an interposer having a first surface and an opposing second surface, wherein the interposer includes conductive pathways through a dielectric material, and wherein the conductive pathways include lines and vias; a first die coupled to the first surface of the interposer by a first hybrid bonding region having a first pitch between 0.1 microns and 10 microns; and a second die coupled to the second surface of the interposer by a second hybrid bonding region having a second pitch different from the first pitch.
2. The assembly of claim 1, wherein the second pitch is between 10 microns and 200 microns.
3. The assembly of claim 1, wherein a thickness of the interposer is between 5 microns and 30 microns.
4. The assembly of claim 1, wherein a thickness of the interposer is between 20 microns and 100 microns.
5. The assembly of claim 1, wherein the dielectric material of the interposer includes an organic dielectric.
6. The assembly of claim 1, wherein a material of the conductive pathways includes copper, silver, nickel, gold, aluminum, or alloys thereof.
7. The assembly of claim 1, further comprising: a third die coupled to the second surface of the interposer by a third hybrid bonding region having a third pitch different from the first pitch and the second pitch.
8. The assembly of claim 7, wherein the third pitch is between 10 microns and 200 microns.
9. The assembly of claim 7, further comprising: a mold material around the second die and the third die.
10. The assembly of claim 9, further comprising: a heat transfer structure on a top surface of the second die, the third die, and the mold material.
11. The assembly of claim 1, wherein the first die has a first surface and an opposing second surface and the interposer is coupled to the second surface of the first die, and the assembly further comprising: a package substrate coupled to the first surface of the first die by hybrid bonding region having a fourth pitch between 0.1 microns and 20 microns.
12. An assembly, comprising: an interposer, comprising: a first substrate layer having first through-substrate vias (TSVs); and a second substrate layer on the first substrate layer having second TSVs;

a first die coupled to first substrate layer of the interposer by a first hybrid bonding region having a first pitch between 0.1 microns and 10 microns; and a second die coupled to the second substrate layer of the interposer by a second hybrid bonding region having a second pitch different from the first pitch.

13. The assembly of claim 12, wherein the second pitch is between 10 microns and 200 microns.

14. The assembly of claim 12, wherein the interposer further comprises: a routing layer between the first substrate layer and the second substrate layer.

15. The assembly of claim 14, wherein the first substrate layer is coupled to the routing layer by a third hybrid bonding region having a third pitch different from the first pitch and the second pitch.

16. An integrated circuit (IC) package, comprising: an interposer, comprising: a first substrate layer having first through-substrate vias (TSVs); a second substrate layer on the first substrate layer having second TSVs; a routing layer on the second substrate layer; and a third substrate layer on the routing layer having third TSVs; a first die coupled to the first substrate layer of the interposer by a first hybrid bonding region having a first pitch; and a second die coupled to the third substrate layer of the interposer by a second hybrid bonding region having a second pitch different from the first pitch.

17. The IC package of claim 16, wherein the first pitch is between 0.1 microns and 10 microns and the second pitch is between 10 microns and 200 microns.

18. The IC package of claim 16, further comprising: a third die coupled to the third substrate layer of the interposer by a third hybrid bonding region having a third pitch different from the first pitch and the second pitch.

19. The IC package of claim 16, wherein the first substrate layer is coupled to the second substrate layer by a fourth hybrid bonding region having a fourth pitch different from the first pitch and the second pitch.

20. The IC package of claim 16, wherein the second substrate layer is coupled to the routing layer by a fifth hybrid bonding region having a fifth pitch different from the first pitch and the second pitch.
